Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *WXIMPACTBENCH* | ***Context:*** *Argentine Government Disaster Reporting Reliability — Policy Staff*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The deployment's primary decision-support output — reliability/credibility scoring of disaster reports — is structurally absent from the benchmark output space, which produces only six binary impact labels [Q25, Q33, Q117] and ranking lists [Q36]. The user envisions credibility as a downstream layer aggregating model outputs with external metadata [A3], so this gap is partially mitigated by deployment design. However, even within the impact-labeling output, the binary schema with no sub-labels limits granularity for Argentine policy needs (villa miseria vs. formal infrastructure; soybean/maize vs. Canadian grain framing). Annotation guidelines emphasize 'direct and immediate effects' and 'explicit references' [Q118, Q147], excluding inferred or slow-onset impacts — a meaningful constraint for ENSO-driven Pampa drought assessment. Empirically, dataset analysis identifies labeling inconsistencies on positive-content chunks (1998 Quebec ice storm with 1M+ without power labeled all-zero [DATASET-D17]; 1881 storm with explicit casualties labeled human_health=0 [DATASET-D27]) that suggest the binary output decisions may not be fully reliable. Output ontology is HIGH-priority per elicitation; score reflects the structural absence of the credibility dimension and granularity constraints, partially offset by user's deliberate design to handle credibility downstream.

ID	Evidence
Q25	"Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)
Q33	"Specifically, given an article sample xt corresponding to six ground-truth impacts Yt = {y^i_t}^6_i=1 with binary labels y^i_t ∈ {0, 1}, the evaluated model M is required to maximize the probability of the predicated impact Ŷ_t = {ŷ^i_t}^6_i=1 towards ground-truth." (p.5)
Q117	"The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)
Q36	"We formulate the ranking-based QA task by prompting the models to identify the likelihood of each article containing the correct answer from a candidate pool." (p.5)
Q118	"The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)
Q147	"Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Strengths

- User explicitly designed deployment to use binary labels as upstream features for downstream credibility aggregation [A3] — output schema is functionally compatible
- Detailed operational criteria provided for each label [Q141–Q146] enable consistent interpretation
- Multi-label simultaneous output [Q57] supports reports spanning multiple impact dimensions

- Six-category schema accepted as broadly applicable to Argentine disasters per elicitation [A1]

ID	Evidence
Q141	“Infrastructural Impact: Classify as 'true' if the text mentions any damage or disruption to physical infrastructure and essential services. This includes structural damage to buildings, roads, or bridges; any disruptions to transportation systems such as railway cancellations or road closures; interruptions to public utilities including power and water supply; any failures in communication networks; or damage to industrial facilities. Consider only explicit mentions of physical damage or service disruptions in your classification.” (p.20)
Q142	“Agricultural Impact: Classify as 'true' if the text mentions any weather-related effects on farming and livestock management operations. This includes yield variations in crops and animal products; direct damage to crops, timber resources, or livestock; modifications to agricultural practices or schedules; disruptions to food production or supply chains; impacts on farming equipment and resources; or effects on agricultural inputs including soil conditions, water availability for farming, and essential materials such as seedlings, fertilizers, or animal feed.” (p.20)
Q143	“Ecological Impact: Classify as 'true' if the text mentions any effects on natural environments and ecosystems. This includes alterations to local environments and biodiversity; impacts on wildlife populations and behavior patterns; effects on non-agricultural plant life and vegetation; modifications to natural habitats including water bodies, forests, and wetlands; changes in hydrological systems such as river levels and lake conditions; or impacts on urban plant life.” (p.20)
Q144	“Financial Impact: Classify as 'true' if the text explicitly mentions economic consequences of weather events. This includes direct monetary losses; business disruptions or closures requiring financial intervention; market price fluctuations or demand changes for specific goods; impacts on tourism and local economic activities; or insurance claims or economic relief measures. Focus only on explicit mentions of financial losses or fluctuations.” (p.20)
Q145	“Human Health Impact: Classify as 'true' if the text mentions physical or mental health effects of weather events on populations. This includes direct injuries or fatalities (including cases where zero or more casualties are explicitly mentioned); elevated risks of weather-related or secondary illnesses; mental health consequences such as stress or anxiety; impacts on healthcare service accessibility; or long-term health implications.” (p.20)
Q146	“Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities.” (p.20)
Q57	“Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Six output labels (Infrastructural, Political, Financial, Ecological, Agricultural, Human Health) [Q25] are regionally relevant per elicitation [A1] and confirmed empirically present [DATASET-D14, D16, D18, D21].

OO-2: Missing categories: (a) credibility / source-reliability / signal-to-noise dimension — the deployment's core decision output, addressed downstream per A3; (b) sub-labels for villa miseria infrastructure vs. formal infrastructure; (c) sub-labels for slow-onset ecological/agricultural impacts (ENSO drought) — guidelines exclude inferred consequences [Q118, Q147]; (d) misinformation likelihood scoring.

OO-3: Categories themselves do not encode non-Argentine values per elicitation [A4]; however, Political Impact criterion [Q140, Q146] frames responses around 'governmental and policy responses' in a Canadian institutional frame that may not capture Argentine partisan polarization or villa miseria framing. The 'direct and immediate' framing [Q118, Q147] systematically excludes the slow-onset and politically-instrumentalized framings common in Argentine disaster discourse.

OO-4: Stakeholder-driven taxonomy redesign is recommended for sub-label addition (informal settlement vulnerability, agricultural commodity-specific loss, partisan-framing political impact). User indicates failure analysis on Argentine data could inform future sub-labeling [A4].

OO-5: Structural validity is harmed by absence of credibility/reliability output (the deployment's primary need) and absence of sub-labels for Argentine-specific granularity. Empirical labeling inconsistencies [DATASET-D17, D22, D27] additionally raise concern about the structural reliability of the binary outputs.

ID	Evidence
Q25	"Six vulnerability-related disruptive weather impacts are defined as the labeling categories, including Infrastructural, Political, Financial, Ecological, Agricultural, and Human Health." (p.4)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D16	Aid Laurent said that there were mountains of snow all over the city. Already $30,000 has been expended in removing it... Seven hundred and fifty men and 450 carters were employed... — 1887 Montreal snow removal debate in city council; all three labeled labels are plausible
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D21	hundreds of houses, schools and other buildings sustained damage when one of the twin peaks of the volcano exploded. At least 20 people sustained minor injuries... The most powerful earthquake to strike western Washington state in 30 years injured four people — 1999 global disaster roundup (Indonesia, Washington state, NZ, Romania); labeled positive
Q117	"The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts." (p.16)
Q118	"The guidelines emphasize focusing on direct and immediate effects, ensuring that classifications are based solely on explicit references within the text." (p.16)
Q147	"Return 'false' for any impact category that is either not present in the text or not related to weather events. Base classifications on explicit mentions in the text. Focus on direct impacts rather than implications. Consider immediate and direct effects." (p.20)
Q140	"Political Impact: Evaluates governmental and policy responses to weather events. Includes government decision-making and policy changes; shifts in public opinion or political discourse; effects on electoral processes; international aid and relations; and debates on disaster preparedness and response. Both direct political reactions and policy implications should be analyzed." (p.19)
Q146	"Political Impact: Classify as 'true' if the text mentions governmental and policy responses to weather events. This includes government decision-making and policy modifications in response to events; changes in public opinion or political discourse; effects on electoral processes or outcomes; international relations and aid responses; or debates surrounding disaster preparedness and response capabilities." (p.20)
D17	PETER MARTIN, GAZETTE Workers from the Connecticut Light and Power Co were working to reconnect power lines on Oxford Ave in Notre Dame de Grace... Temperatures plummet; today's high minus-15C... more than one million people struggle on without power or heat — 1998 Quebec ice storm coverage describing power outages and extreme cold; labeled all-zero — potential annotation gap
D22	Almost 4 million people across China had been cut off by flood waters, 810,000 homes have collapsed and 2.8 million homes have been damaged in eight provinces... Japan struck at young and old yesterday, killing a schoolgirl and an 85-year-old woman — 1996 China Yangtze floods + Japan E. coli outbreak in same chunk; infrastractural=1 but human_health=0 despite explicit casualties
D27	A poor lady passenger was dashed to leeward and had her skull fractured. She was landed this morning in a dying state with a coffined child that succumbed last night to its sufferings. — 1881 Atlantic storm with explicit casualties (skull fracture, child death) labeled all-zero

Information Gaps

- Whether Argentine policy users would accept 'direct and immediate' criterion or require slow-onset assessment for ENSO drought
- Specification of downstream credibility aggregation pipeline that A3 references

Requires Expert Verification

- Whether the labeling inconsistencies flagged in dataset analysis [D17, D22, D27] constitute systematic errors or chunk-level re-annotation per Q47/Q48

Remediation

Gap 1: No native credibility/reliability output and no sub-labels for Argentine-specific granularity [Q117, A3]

Recommendation: Design and document the downstream credibility aggregation layer that A3 envisions, specifying Argentine source-metadata schema (outlet reach, account verification, AFE/SINAGIR/SMN actor recognition). Add sub-labels for villa miseria infrastructure, agricultural commodity specificity (soybean/maize/wheat), and partisan-framing political impact. Treat WXImpactBench outputs as one of several upstream features.

ID	Evidence
Q117	“The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts.” (p.16)